Improved Accuracy in Molecular Transport Simulations Utilizing Multispecies Decomposition with QCM-Derived Sticking Coefficients

Jacob Meyer,[1] Lubos Brieda,[2,3] and Joseph Wang*[1]
[1)] *University of Southern California, Los Angeles, CA 90089*
[2)] *Particle in Cell Consulting LLC, Westlake Village, CA 91362*
[3)] *California State Polytechnic University, San Luis Obispo, CA 93407*

(*Electronic mail: josephjw@usc.edu)

(Dated: 31 October 2025)

This paper applies a multi-gaussian decomposition technique to derive multi-species sticking coefficients from Thermogravimetric Analysis (TGA) data for Spacecraft Contamination and analyzes the performance improvement of the multi-species sticking coefficient approach for a multi-QCM experimental setup with one out-of-sight QCM. This paper also applies a technique to analytically derive the physics-based activation energies corresponding to the multi-gaussian decompositions rather than only relying on empirical sticking coefficient fits, and the results of the two approaches were also compared. The analysis was performed on data collected during a previous joint Blue Origin USC collaboration and compared to the previous state of the art approach which only utilized single-species Quartz Crystal Microbalance (QCM)-derived sticking coefficients for modeling gray-body view factors in spacecraft contamination, with the goal being to assess the improvements gained by utilizing a multi-species approach over the previous single-species approach. When utilizing the single-species sticking coefficient model, the simulations were found to greatly under-predict contaminant collection on the second QCM compared to experiments, but the multi-species decomposition method yielded much better agreement between the simulation and experiments. The empirical multi-species sticking coefficient fits also proved to be more accurate than the analytical multi-species activation energy approach.

I. BACKGROUND

Materials exposed to the vacuum environment undergo outgassing, a process in which volatile molecules embedded within the native material diffuse to the surface and desorb into the external environment. Outgassing is primarily associated with the release of small hydrocarbon precusor molecules that have undergone only a limited polymerization during the production of organic compounds. Other molecules, such as water, can also be of interest especially in the case of cryogenic applications. Outgassing is of concern to designers of space missions, since the deposition of outgassed contaminants can lead to transmission loss in optical instruments or a change in emissivity in thermal control surfaces. The thickness of molecular contamination films that can be expected to deposit during the mission duration can be estimated using numerical mass transport analyses[1–4]. These codes utilize ray or particle tracing coupled with deposition and re-emission models to calculate the gray-body viewfactor between contaminant sources and areas of concern. Assigning models governing the surface behavior is not trivial. The time that a molecule is expected to remain on a surface before desorbing due to random vibrational events is called residence time[5,6]:

$$\tau = \tau_0 \exp\left(\frac{E_a}{RT_s}\right) \tag{1}$$

where τ_0 is assumed to be $\approx 10^{-13}$ seconds[5], E_a is the activation energy for this process, T_s is the surface temperature, and $R \approx 1.987$ cal/mol/K is the universal gas constant. The residence time can be seen to be strongly dependent on the E_a/T ratio. For example, molecules with activation energy of 11 kcal / mol has a residence time of 10^{17} s on a surface at 77 K, which is on the order of the age of the universe, but resides for less than a nanosecond at room temperatures. Therefore, the first surface can be assumed to act as a molecular sink, while the second one acts as a mirror, possibly reflecting such molecules to regions with no direct of line of sight to the contaminant source.

The contaminant population is composed of a multitude of chemical species of varying activation energies. While mass spectrometers (such as residual gas analyzers, or RGAs) could conceptually provide an insight into the chemical composition, the environment encountered in vacuum chambers during system-level thermal vacuum test is dominated by atmospheric gases such as nitrogen, oxygen, or water, with trace hydrocarbons generally lost in the noise. This issue is further compounded by a fixed mass limit (such as 200 Da) to which the RGA is able to scan. For this reason, molecular outgassing tends to be characterized in-situ using thermally-controlled quartz-crystal microbalances (TQCMs). These devices record the frequency of a small quartz crystal disk exposed to the vacuum environment. Crystal frequency is proportional to mass and QCMs thus directly measure condensation rate at the temperature corresponding to the crystal temperature[7]. By taking into account the view factor to a source, this deposition rate can be correlated to the material outgassing rate. The deposited film is removed whenever the deposited mass reaches a threshold value by increasing the crystal temperature to $> 80\ ^\circ$C. The crystal temperature can be programmed to increase instantaneously or at a constant rate to perform a QCM thermogravimetric analysis, or QTGA, to obtain desorption rate as a function of temperature. While a QTGA does not provide any direct insight into the chemical species present on the surface, it helps to identify the number of "virtual species" present on the surface, each with a characteristic surface deposition temperature.

A common approach for modeling mass transport is to assume that the molecular population is composed of a single contaminant with a temperature varying sticking coefficient, $\alpha_s = \alpha_s(T)$ derived from the QTGA, as demonstrated in various prior works[3,8]. This single species model can however be expected to misrepresent gray body mass transport in configurations with multiple surface bounces. This was in fact demonstrated in our past work[9], in which we used a TQCM with a direct line of sight to an outgassing sample to obtain $\alpha_s(T)$. The sticking coefficient was then used in an in-house simulation code[?] to predict deposition to a second QCM facing vacuum chamber wall, and hence having no direct line of sight to the outgassing sample as visualized in Figure 1. The experiment was repeated at University of Southern California (USC) and Blue Origin (BO) facilities, with both configurations producing comparable trends. The discrepancy arises from the single–species model unable to capture the physical behavior in which the contaminant plume impinging on a surface bifurcates into a population that adheres to it and a population that reflects. In the case of multiple wall impacts in an isothermal environment, as was the case with the mass transport in the vacuum chamber, a multi-species model leads to the contaminant species with deposition temperature greater than the wall temperature to stick on the first wall impact. The rest of the contaminant population then continues to bounce around the chamber until finding its way to the colder QCM, where it deposits. The single species model, on the other hand, leads to a concentration reductions relative to $\alpha_s(T_{wall})$ on each wall impact.

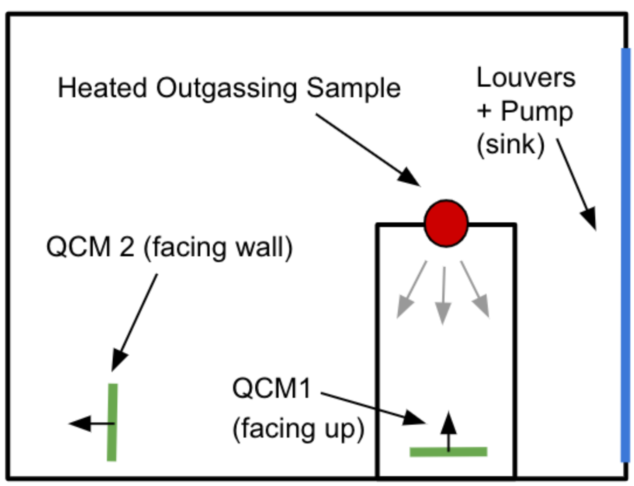


FIG. 1: Schematic drawing of experimental setup used to characterize gray-body mass transport. QCM1 has a direct line of sight to an outgassing sample, while QCM2 is facing the chamber wall.

An alternative approach involves utilizing Gaussian decomposition to fit multiple virtual species to the QTGA data has been demonstrated in[10? –15]. To the best of our understanding, this multispecies has not yet been validated in context of predicting gray-body contaminant mass transport. In this letter, we apply the multi-species approach to data from our prior work[9]. We begin by describing the process of fitting virtual species to the QTGA data. We subsequently present results from mass transport analyses with a varying number of virtual species. In this work, we demonstrate that the multi-species approach results in an improvement of several orders of magnitude compared to the legacy single-species sticking coefficient model.

TABLE I: Summary of experimental and numerical deposition rates to QCM2 for the single species model[9]. The experimental and numerical columns represent the ratio of the mass accumulated on QCM1 divided by that on QCM2.

Configuration	Experimental	Numerical	Error (%)
BO harness	1.0	8.2	720
BO cable, warm wall	1.45	25.6	1666
BO cable, cold wall	5.0	194.8	3796

II. METHODS

The multispecies sticking model assumes that the TGA signal is a mixture of multiple desorption events - that is, that the signal is the sum of several individual species, each assumed to be well represented by individual gaussians in the TGA. To apply the multispecies sticking coefficient model to our work, we imported the QCM 1 QTGA data into the commercial scientific analysis software "OriginLab" and used the "peak deconvolution" feature to make the gaussian fits as seen in Figure 2[16]. We did this for the QCM 1 TGA data for all three cases presented in the Blue Origin / USC paper, and for each case, we decomposed the TGA data into 2 and 3 gaussian cases to further investigate the effect of number of gaussians on the results, with a 4-gaussian fit also being explored for the BO cable, warm wall case as well since the best fit for that case consisted of four gaussians when analyzing it with the Peak Deconvolution tool in OriginLab.

The QCMs in Brieda et. al 2022 collected material for approximately an hour or slightly less in each of these cases before the crystal had to be baked off due to the high outgassing rates. However, since the QCMs only were able to heat up at approximately 3°C/min during the TGA process, the TGAs themselves took a significant fraction - over half of the collection time, to finish. Due to this, these TGAs included species which were not present at the beginning of the TGA which we needed to account for in our final analysis.

To do this, "blank" TGAs were taken with QCM2 for each of the three cases, where two TGAs were taken in rapid succession - with the second one being used as a "blank" to characterize the ambient material in the chamber during the TGA phase. This "blank" would only include the material that accumulated during the TGA phase, since it would begin with no material deposited initially. For each of the three cases, the blank was subtracted from the original "raw" QCM 2 TGA curve in order to yield the true, corrected TGA curve which only includes the material collected before, but not during each TGA. This correction process is shown in Figure 3.

We also included a fourth case based on the Blue Origin /

USC data not included in the Brieda et. al 2022 paper, which uses a "hot to cold" TGA approach that gives a clean signal without the need for a separate blank, even with the TGA time being on the order of tens of minutes. The "raw" QCM1 and QCM2 data in this case are shown in 4a, and the "true" data for these cases after processing are shown in figure 4b. The "raw" hot-to-cold data was converted to the "true" TGA curves for QCM1 and QCM2 by dividing the raw data into equal-sized bins and then - from left to right - setting each bin's new midpoint y-value equal to itself minus its neighbor to the right. This gives us the "true" TGA signal and subtracts the signal since each bin includes the contaminants that deposit at that specific temperature plus all of the contaminants that deposited at all higher temperatures (including noise). Thus, subtracting each bin from its neighbor to the left gives the true TGA signal at that temperature since it subtracts background contamination and noise at the same time.

Similar to the data from the three cases in Brieda et. al 2022, the true TGA curves from Figure 4b here were decomposed in OriginLab using a 6-Gausisan multispecies fit. Six gaussisns were used instead of 3 or 4 since that was determined by OriginLab to be the "best fit" versus any number of gaussians for that case.

Once the gaussian fits were made for each case, the sticking coefficients for the individual gaussians at the range of temperatures of interest were all determined via the standard gaussian cumulative distribution function shown below:

$$F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{2}\left[1+\operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right] \tag{2}$$

To calculate sticking coefficients, x corresponds to the temperature at any given point, μ is the temperature around which the gaussian is centered, and σ is the standard deviation of each of the gaussians. The calculated value of $F(x)$ corresponds to the sticking coefficients for each gaussian across the domain in our case.

The sticking coefficients, relative outgassing mass flux of all of the Gaussians (determined by the areas under the Gaussians relative to each other), and the CAD model and wall temperatures of each case were input into the previously mentioned Contamination Transport Simulation Program (CTSP) Code to numerically simulate each case.

Lastly, one last set of additional activation energy-based CTSP simulations was also performed for the hot-to-cold TGA case, to see if a more physics-based approach could be possible rather than simply making arbitrary Gaussian and sticking coefficient fits.

To do this, we reused the 6 Gaussians from the hot-to-cold multispecies sticking coefficient case and attempted to extract a physical activation energy for each of them. This was done via plugging the Arrhenius equation into an exponential cumulative distribution function as shown below.

$$F(x) = 1 - \exp(-\lambda\delta t) \tag{3}$$

Where $\lambda =$ the Arrhenius equation, and the final result is:

$$C_{des} = 1 - e^{-\frac{dt}{\tau_0}e^{\frac{E_a}{RT_S}}} \tag{4}$$

When extracting activation energies via this method - we approached this with two different methods and compared the results to see which (if either) was better. Both methods involved plotting the Gaussian cdf and the above arrhenius-based exponential cdf function for each gaussian together and adjusting the activation energy of the C_{des} equation until the two curves lined up. In the first version of these simulations that we ran, we kept the pre-exponential factor constant at $\tau_0 = 10^{-13}$ seconds. In this case, we adjusted the activation energies until the center of both of the cdf curves intersected. In the second case, we adjusted both activation energy and the pre-exponential factor to get a better graphical "fit" between the two curves. Changing the activation energy only shifts the curve to the left or the right, but changing τ_0 adjusts the slope of the curve as well, and allowed us to create better visual "fits" between the Gaussian CDF and the Arhennius-based exponential CDF. Some of the values of τ_0 that we ended up using were unphysical (e.g. on the order of $\tau_0 = 10^{-45}$) in order to get the curves to match, but the results actually ended up being better than when keeping $\tau_0 = 10^{-13}$. In both cases, the E_a that was extracted for each gaussian corresponded to the best visual fit between the two equations on a plot (with only case 2 fitting exactly due to the ability to adjust tau and thus the slope).

Both cases were also plugged into CTSP, but instead of using sticking coefficients corresponding to each gaussian, the activation energies which corresponded to each Gaussian that were extracted via the process above were input into the code, which is able to simulation gray body transport in terms of activation energies, not just sticking coefficients. The results were then compared to the multi-species sticking coefficient approach for the hot-to-cold TGA case.

III. RESULTS & DISCUSSION

As can be seen in Table 2, the multispecies decomposition resulted in more accurate predictions of molecular deposition on the second QCM compared to the previous unitary sticking coefficient model, with predictions generally improving as the number of Gaussians seems to approach the number of (significant) species which appear to be present in the data. However, as already mentioned, we should strongly emphasize that the number and shape of Gaussians and/or virtual species present does not necessarily represent an exact one-to-one correspondence with real chemical species but rather is the current best empirical fit per the Peak Deconvolution Methods used.

Similarly, the "hot to cold" TGA case shows strong agreement with the multi-species coefficient approach, with the experimental data an error of only 5.45% compared to the "best fit" 6-gaussian case in CTSP. However, when the two analytical activation energy approaches were tested on the same Gaussians, the results were far less accurate than the ESA

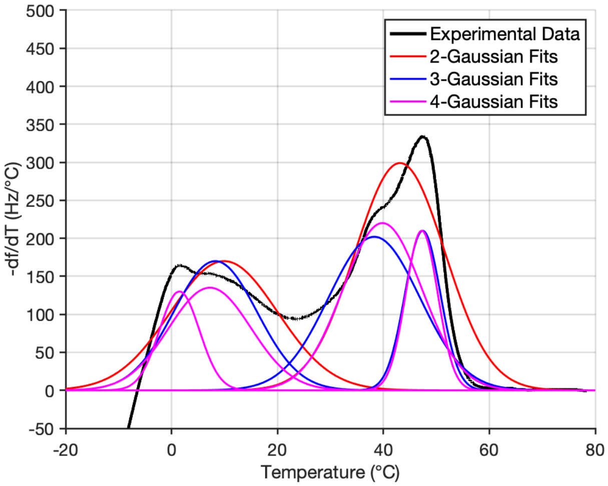


FIG. 2: Multi-Gaussian Fits for QCM1's TGA from the "Unbaked Cable, Warm Shroud" Case from the BO/USC Paper, with the 2-gaussian (red), 3-gaussian (blue), and 4-gaussian (pink) gaussian decomposition fits shown

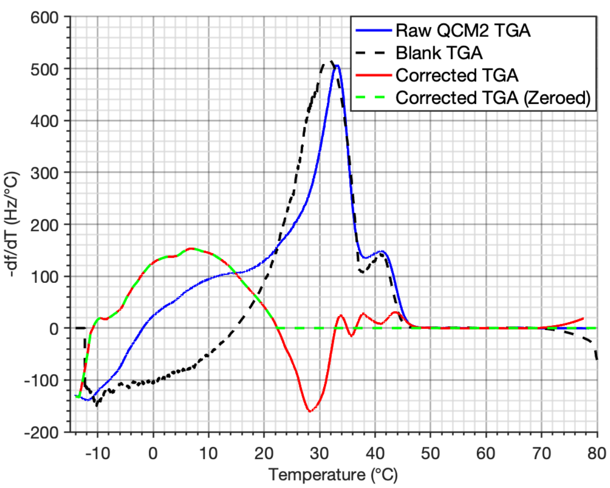
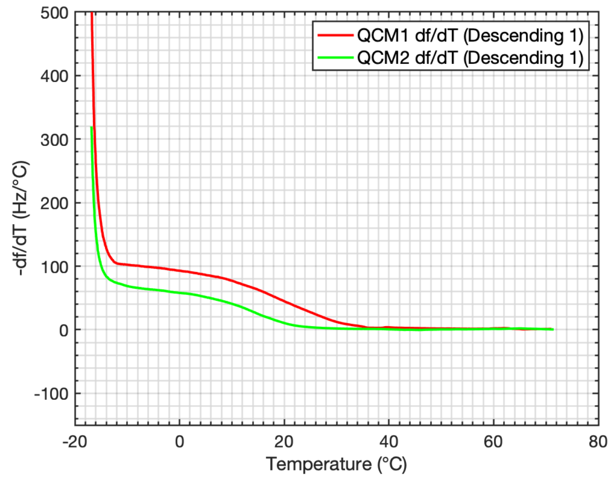


FIG. 3: Original (Blue) and Corrected (Red / Green) QCM2 Experimental TGA Signal after Blank Subtraction

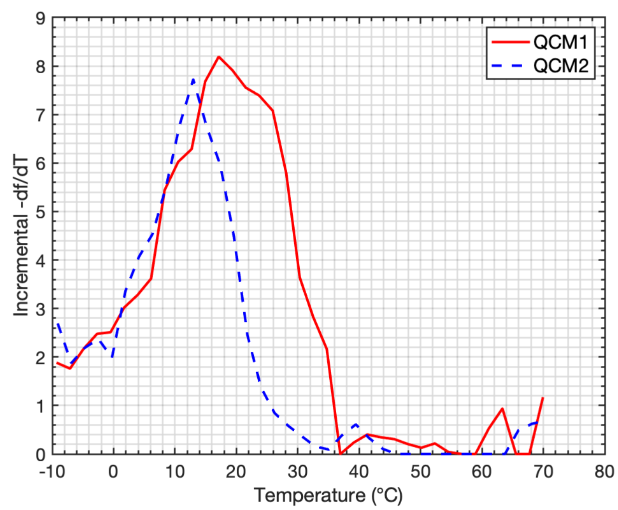
multispecies sticking coefficient approach. The first approach, with a fixed τ_0, was less accurate than the second approach where both E_a and τ_0 were adjusted, with τ_0 being adjusted to artificially high levels well above the standard 10^{-13} in the latter case.

IV. CONCLUSIONS

This paper demonstrated the use of normal-distribution fits to recover multiple virtual contaminant species from a QTGA data. The corresponding activation energies were then used in a particle-tracing mass transport simulation program to predict molecular deposition onto a secondary QCM with no direct line of sight to the outgassing source. Numerical results



(a) RAW "Hot-to-Cold" TGAs for QCMs 1 and 2



(b) Corrected "Hot-To-Cold" TGAs after Discretization and Sequential Neighbor Bin Subtraction

FIG. 4: "Hot-To-Cold" TGAs.

were compared to the previously-reported experiment, and also to numerical predictions obtained with a legacy single-species sticking coefficient approach. We find that increasing the number of virtual species leads to a general improvement in results, with relative error reducing by up to 3 order of magnitude. Ongoing Molecular Dynamics (MD) studies and other future work is being undertaken at USC to further investigate and refine these models, including effect of source-target material, dynamic restructuring, and other possible phenomena to further improve our understanding and modeling of multi-species sticking coefficients[17].

REFERENCES

[1] Z. Jiao, L. Jiang, J. Sun, J. Huang, and Y. Zhu, "Outgassing environment of spacecraft: An overview," in *IOP Conference Series: Materials Science and Engineering*, Vol. 611 (2019) p. 012071.

TABLE II: Comparison of Experimental and Numerical Multispecies Deposition Rate Reductions, and Error Relative to the Experiment for the three **Cold-to-Hot TGA** cases.

Test	Experimental	Numerical	Error (%)
BO harness			
1-Gaussian	1.0	8.7	768
2-Gaussian	1.0	2.1	114
3-Gaussian	1.0	2.4	142
BO cable, warm wall			
1-Gaussian	1.45	45.9	3062
2-Gaussian	1.45	2.4	62
3-Gaussian	1.45	2.1	43
4-Gaussian	1.45	1.39	4.3
BO cable, cold wall			
1-Gaussian	5.0	153.3	2966
2-Gaussian	5.0	46.8	837
3-Gaussian	5.0	28.7	474

TABLE III: Comparison of Experimental and Numerical Results for the **Hot-to-Cold TGA** validation case.

Test	Experimental	Numerical	Error (%)
BO Hot-To-Cold TGA			
6-Gaussian Sticking Coeff.	1.65	1.56	5.5
E_a Fits (τ_0=10^{-12})	1.65	42.9	2502
E_a Fits (Custom τ_0)	1.65	11.9	620

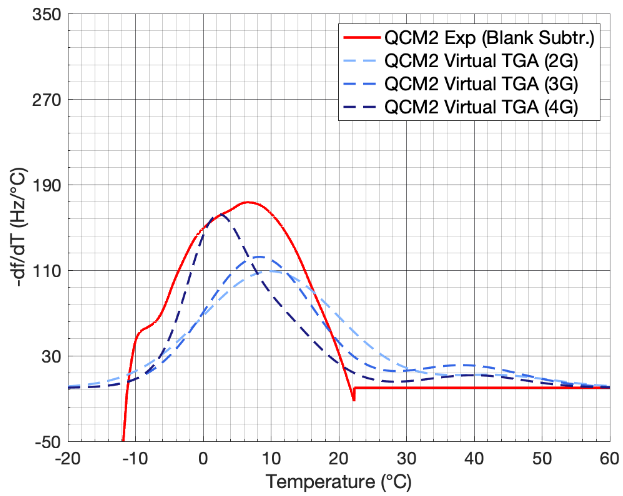


FIG. 5: Experimental QCM2 TGA vs Predicted QCM2 TGAs Based on simulations for each set of Gaussian fits (BO unbaked cable, warm shroud case)

[2] L. Brieda, M. Laugham, M. Woronowicz, K. Henderson-Nelson, C. May, and E. Woolridge, "Numerical study of water ice and molecular contamination build up during jwst deployment," in *Proc. SPIE 12224, Space Systems Contamination: Prediction, Control, and Performance* (2022) p. 1222403.

[3] E. M. Petro, D. W. Hughes, M. S. Secunda, P. T. Chen, J. R. Morrissey, and C. A. Riegle, "Maven contamination venting and outgassing analysis," Tech. Rep. (NASA Goddard Space Flight Center, 2016) nTRS Accession: 20160010595.

[4] L. Brieda, "Numerical model for molecular and particulate contamination transport," Journal of Spacecraft and Rockets **56**, 485–497 (2019).

[5] A. C. Tribble, *Fundamentals of Contamination Control* (SPIE Press, 2000).

[6] J. Kohout, "Modified arrhenius equation in materials science, chemistry, and biology," Molecules **26**, 7162 (2021).

[7] ASTM International, "Standard test method for contamination outgassing characteristics of spacecraft materials (e 1559-93)," Annual Book of ASTM Standards (1993).

[8] J. M. Alred, B. D. Moore, S. Susca, V. Ricchiuti, J. M. Rocca, and C. E. Soares, "Modeling contaminant outgassing and free molecular transport processes for the cryogenic spherex observatory," in *AIP Conference Proceedings*, Vol. 2996 (2024).

[9] L. Brieda, E. Helou, T. Zhao, J. Wang, D. Aranda, V. Green, and S. Iqbal, "Experimental investigation of qcm-derived sticking coefficients for use in molecular transport simulations," in *Proc. SPIE 12224, Systems Contamination: Prediction, Control, and Performance* (2022) p. 122240O.

[10] A. Suliga, O. Ergincan, and R. Rampini, "A new dynamic outgassing procedure to predict re-emission and deposition values of contaminants," in *Proc. SPIE 11489, Systems Contamination: Prediction, Control, and Performance* (2020) p. 114890E.

[11] A. Suliga, O. Ergincan, and R. Rampini, "Modeling of spacecraft outgassed contamination levels by thermogravimetric analysis," Journal of Spacecraft and Rockets **58** (2021), 10.2514/1.A35020.

[12] D. Lansade, J. F. Roussel, J. Alred, A. Wong, J. Anderson, C. Soares, D. Faye, and G. Rioland, "Separation of contaminant species by tga/ms in european and u.s. approaches to outgassing," in *Proc. SPIE 12258, Systems Contamination: Prediction, Control, and Performance* (2022) p. 122240M.

[13] D. Lansade, J. Roussel, D. Faye, G. Rioland, A. Suliga, and O. Ergincan, "Contamination level prediction: Progress in species separation by tga/ms," in *IOP Conference Series: Materials Science and Engineering*, Vol. 1287 (2023) p. 012022.

[14] J. M. Alred, M. G. Martina, W. A. Hoey, A. T. Wong, L. M. White, P. A. Boeder, S. A. Cofer, A. Dias-Ribeiro, and C. E. Soares, "Predicting terrestrial contamination of the mars 2020 sample caching system with novel multispecies outgassing and transport models," in *Proc. SPIE 11489, Systems Contamination: Prediction, Control, and Performance* (2020) p. 114890H.

[15] M. C. Fong and A. L. Lee, "A time-dependent sticking coefficient model for multi-species spacecraft material," in *38th AIAA Thermophysics Conference* (2005).

[16] R. May and K. Stevenson, "Software review of origin 8," Journal of the American Chemical Society **131**, 872 (2009).

[17] J. Meyer, L. Brieda, and J. Wang, "Molecular dynamics simulations of heterogeneous molecular contaminant films," in *Proc. SPIE 13628, Space Systems Contamination: Prediction, Control, and Performance*, Vol. 13628 (2025).

[18] Y. Abul-Huda, R. Darragh, S. MacKenzie, R. Lorenz, and Z. Turtle, "The dragonfly mission: Contamination transport during entry and descent of titan," in *Proc. SPIE 13628, Space Systems Contamination: Prediction, Control, and Performance* (2025) p. 1362807.